Map-Reduce Multi-way Join

* Indicates required question

1. Email *

2. Roll number *

3. Password *

Question

We wish to take the join R(A,B) |><| S(B,C) |><| T(A,C) as a single MapReduce process, in a way that minimizes the communication cost. We shall use 512 Reduce tasks, and the sizes of relations R, S, and T are 2^20 = 1,048,576, 2^17 = 131,072, and 2^14 = 16,384, respectively. Compute the number of buckets into which each of the attributes A, B, and C are to be hashed. Then, determine the number of times each tuple of R, S, and T is replicated by the Map function. Find the correct statements.

4. 1 point

Check all that apply.

- [ ] Tuples of R are sent to one Reduce task.
- [ ] Tuples of T are sent to 64 Reduce tasks.
- [ ] Tuples of S are sent to 64 Reduce tasks.
- [ ] Tuples of S are sent to 8 Reduce tasks.
- [ ] Tuples of R are sent to 8 Reduce tasks.